Radio Frequency Identification (RFID) in Libraries

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ABSTRACT

This study was carried out at a time when technological developments had revolutionized life. The adoption is just but inevitable in all aspects of life. The library, which is a growing organism is no exception to this and has not been left out in the adoption of technology. The study investigated the impact of RFID in the library, how it changed its operations and the challenges that experienced in the adoption and use. The study specifically focused on the impact of RFID technology to the library basic housekeeping activities namely: security management, shelving, circulation, information resource retrieval as well as the inconveniences and the challenges arising from the adoption of RFID technology in the CUEA library. The objectives of the study were to establish the current situation in libraries, to establish the loopholes, to establish if RFID offers any solutions to the loopholes and to establish the challenges associated with the adoption of RFID in libraries.

Data was collected from two categories of respondents namely: library users and library staff using two sets of questionnaires one for each category. It was the analysed using by use of percentages, tables, pie-charts and bar graphs. The finding of this study identified the impact of RFID in libraries and recommended some issues for immediate action by libraries wishing to adopt RFID or even those which have already adopted RFID. These included: the need to train users, the need to train the staff, carrying out a thorough cost-benefits analysis before analysis especially for small libraries with few collections. The study also calls for research in areas with research gaps which included the lack of standards in RFID use in libraries and the challenge of tag collision.

INTRODUCTION

1.1 Introduction

This chapter provides a brief background to the study and tries to explain the problem that the study sought to explain. It also presents statement of the problem, research objectives, and research questions, the significance of the study, scope of the study, assumptions of the study, theoretical framework and definition of terms.

1.2 Background of the study:

Universities throughout the world exist to support and excel in learning, teaching and research amongst the students, staff and the university fraternity in general. To achieve these goals, universities establish libraries which are mandated to collection, processing, ensuring access and disseminating of information (Vasishta & Badhi, 2009) to the university community. A library is an institution of knowledge provision and learning in a university. It provides invaluable services to its clients. Universities have realized that the scope of knowledge is wide and widespread for it to be covered extensively and comprehensively within the confines of a classroom situation however superior the method of instruction may be which has propelled universities to develop libraries for academicians to search for more knowledge.

Libraries have realized the noble role they play in universities and hence have focused on offering their clients better services by emphasizing on the five principles of librarianship as stipulated by S. Ranganathan.

We are living in the information age characterized by technological advancement and information

explosion where libraries have grown with time in the social, cultural, scientific and technological environment in which they operate (Dhiman, 2004). According to Banerjee (1996), libraries all over the world will be in ceaseless transition due to technological advancement, and the library environment will be constantly changing which will place significant demands on the library staff and the management to adopt the new changes.

Library automation has been an area of major concern for university libraries in a bid to serve the growing number of users as well as to excel in their role of supporting teaching, learning and research in the universities. This has helped the library to reduce on costs, save the time of the user, free library staff from some roles and duties allowing them to contribute more meaningfully to the dissemination of information and to improve customer satisfaction as well as service quality. Libraries and other information centres have had to adopt information communication technology (ICT) in order to keep abreast of user expectation and the global changes. The adoption of ICT in libraries has been referred to as library automation (Mohammed, 1991).

Library automation is not new in the library; it has developed over time in the different functions of the library. The period from 1930-1960 marked the beginning of library automation where computers were used to automate library circulation and acquisition while the period between 1960s and 1970s saw improvements in the use of computers and this led to an increase in library automation (Meckler, 1991). Since then, libraries have increasingly continued to automate their major processes and services such as cataloguing, acquisition, and circulation of library information materials (Computer-Based National Information Systems: Technology and Public issues, 1981:8).

Today, barcodes and electromagnetic strips are the technologies currently in use in most of the university libraries in Kenya in automating circulation and security management.

The Catholic University of East Africa (CUEA) library, however, has taken lead in adopting a new technology in library automation referred to as Radio Frequency Identification (RFID) which is said to be an improvement of the barcode and the electromagnetic strip technology. The RFID technology helps to automate a number of major library functions including circulation (self-check-out/ self-check-in, instant check-out/check-in), inventory management, shelving (easily identifying mis-shelved materials, automatic sorting of information materials, enhancing retrieval of information materials from the shelves using a hand-held gadget) and enhancing the security of the library information resources and most importantly it increases the quality of service offered to the users.

According to Kavulya (2014) providing access to information through the use of RFID technology is both a complex and costly process but he says it is a technology that has yielded the library fruits and it's worth adoption by other university libraries in Kenya and all over the world.

This study therefore aimed at analyzing the situation at CUEA library after the adoption of RFID technology in the library basic housekeeping activities such as circulation, shelving, security management and inventory control.

1.3 Statement of the problem:

The current situation in most university libraries in Kenya is not appealing to library client service as well as in enhancing security of library materials. The barcode and electromagnetic strip technology currently in use is not efficient enough as library users and staff realize their inefficiencies and loopholes. It is common to see long queues at the circulation counter especially at peak hours of the day something which is frustrating especially to students who may either be rushing to attend a lecture, or even to catch up with the evening bus. The circulation librarians on

the other hand are frustrated by the barcode readers/scanners which either fail to immediately scan the barcodes or the tedious work of opening the materials and aligning the barcode with the infrared rays from the barcode scanners. Some of the books are too bulky and voluminous for the circulation librarian to keep on handling them each at a time in order to scan the barcode either when users are checking-in or checking-out the materials.

The users are also frustrated when the go to the shelves to retrieve a book which according to the library Online Public Access Catalogue (OPAC) is available in the library shelves only to miss it. Upon consulting with the library staff, the response is as usual, "*from the library OPAC, we have many such copies on the shelves, kindly check properly*". This library staff is unaware that the book has been mis-shelved.

The shelving librarian simply placed the materials on the shelves without following any classification scheme because he/she had to complete another assignment from the head librarian due the following day. He/she is really overworked.

After compiling the library periodical statistics, it is realized that some information materials cannot be accounted for! How is this possible even with the electromagnetic strip security installed and having library security guards at the exit to ensure that no resources leave the library without proper check-out? Some users or staff members have realized the loopholes of electromagnetic strip security, and they simply remove the electromagnetic strip from the information material and they freely walk outside the electromagnetic security gates at the exit without call for alarm, after the materials has been deactivated through removing of the electromagnetic strip.

The library system has quite a number of loopholes now which the management should address which range from poor user service by not saving the time of the users at the circulation desk and at the library shelves when trying to retrieve information materials to not providing them with the

required information materials because they have been stolen by users or even some of the library staff. It becomes quite expensive to replace the stolen information materials. The library pays out a lot of money to the guard at the exit but it still experiences theft of information resources. This study was designed to discuss how the use of RFID technology can be adopted and help to curb this situation in the library and other information centres.

1.4 Research objectives:

The general objectives of this study will be to analyse the use of RFID technology in the library through focusing on the following specific objectives.

a) To access the current situation based on circulation, shelving, inventory and security issues.
b) To access any loopholes in the current situation.
c) To try to establish the possible solutions offered by the application of RFID technology in the library.
d) To identify the challenges that are associated with this new technology.
e) To establish the benefits of this technology in libraries.

1.5 Research questions:

The research questions for the study were:

a) What type of circulation problems exist in university libraries?
b) What challenges does the library staff encounter during shelving of information materials in university libraries?
c) How efficient is the application of RFID technology in the library basic house-keeping activities?

d) Does the adoption of RFID technology lead to better user service and user satisfaction in the libraries?

e) Does the adoption of RFID technology help to reduce the loss of information materials from the library?

1.6 Significance of the study:

The significance of this study is to help understand how developing and adoption of RFID can help to deal with the circulation process, shelving, retrieval and ensuring security and safety of the library materials. It is a study that will also help a university library to help contain its operational and resources replacement costs and to improve the quality of customer service, help to reduce or avoid injuries to staff, help in saving time of the user when checking out materials or even when trying to locate materials on the shelves and help in ensuring that there is maximum utilization of the library staff to increase efficiency. The study will also help an information centre to provide better and quality service to its users in terms of flexibility in returning borrowed information resources by allowing self-check-in on a 24/7 basis.

Costs related to replacing lost items are a prime threat to a library's small budgets and this study aims at studying how the application of RFID technology can be used to solve this problem.

1.7 Scope of the study

A number of authors have published literature on the different aspects of the application and adoption of RFID technology in libraries. The study aimed at identifying these aspects and determining to what extent they affect the library housekeeping activities such as checking in and checking out, shelving, sorting of books, retrieving books from the shelves and ensuring the security of the information materials at the CUEA library

1.8 Assumptions of the Study

The research was based on various assumptions such as low return rate of the questionnaires and not knowing whether it is the intended respondent who responded to the questions, the researcher assumed that the sampled population will cooperate fully by attempting all the questions and that the questionnaire will be filled by the intended respondents. The researcher also assumed that the respondents would give honest and unbiased responses despite the fact that the use of questionnaires as a method of collecting data has been criticized for the respondents not being honest. The researcher also assumed that the respondents participated in the study without any hidden personal motives such as to impress the researcher or to receive certain favours such as being acknowledged in the study.

1.9 Limitation of the study

The researcher experienced the following challenges while undertaking the study:-

- Lack of enough literature review materials. It was costly to research by browsing the information from a cybercafé and going to the library for more information at Kenyatta University.
- Travelling expenses to the CUEA library was also another challenge the researcher experienced. The researcher did not have enough finances to fund the travelling expenses involved.
- Some respondents were uncooperative with some failing to return the questionnaires within the agreed time.

1.10 Definition of operational terms

Library automation- it is the application of computers in the basic activities of the libraries. An automated library is the one which has adopted the use of computers to help in carrying out its activities such as information awareness, user education, acquisition, circulation etc.

Check-in- it is the process of a library receiving back an information material that was lend out to a library user.

Check-out- it is the process of a library allowing a registered library user to borrow an information material usually for use outside the library and return it back after a specified period failure to which a fine is charged.

Self-check-in/out- it is a system which gives the user of an information centre the exclusive rights to borrow and return borrowed information resources without the assistance of a circulation librarian.

Self-check-in/out kiosks/points- these are points (machines) located at the library or busy points such as the airports, bus stops to allow users carry out self-check-out or check in of information resources.

Library client(s) - it is the user of a library or an information centre. These are the students, staff members and the community members allowed by the CUEA to use the library resources. Library clients are also called library patrons or simply patrons.

Library housekeeping activities- these are the basic activities of an information providing centre such as a library which help to ensure that the library meets its core objective of providing access to information resources. The activities involved include acquisition, classification and cataloguing, shelving, weeding out, ensuring security of the materials etc.

Circulation- it is the movement of information resources into and out of the CUEA library. When

materials are moving out of the library, the process should be authorized by the library system and policy in place.

Information age- it is an era (period) where information is viewed as a factor of production and a necessary resource for steering development both at personal and impersonal level.

In the information age, knowledge is attributed to power.

Information explosion- it is a situation where information is being produced at a rate at which libraries and information centres are not able to contain. It has resulted to so much information leaving information providers at a dilemma of which information to provide to users and which one not to.

Respondents – any person(s) (students or librarians form CUEA) who will give any information as may be requested during the study.

Conventional - Something that is adopted widely becoming acceptable to majority and has been in use for quite some time.

Library security systems- these are the measures that have been adopted by the CUEA library to ensure that its collection is not taken out of the library.

Disseminate: It is to share and to distribute research findings within the CUEA university community.

Adoption: It is the decision to implement, take up or use computers as a way of disseminating information and research findings to the users of CUEA.

LITERATURE REVIEW

2.1 Introduction

Literature review is defined by Oso and Onen (2005) as a systematic identification, location and analysis of the documents containing information related to the research problem. According to Bak (2004) literature review should have boundaries as it is impossible to review all there is in a subject and it helps to define and limit the problem the researcher is working on by comparing and contrasting different authors' views on the same topic. Oliver, (2004) argues that the major purpose of writing a literature review is to establish the academic and research area that is important to the subject of the study. Mugenda and Mugenda (1999) suggests that the purpose of literature review is avoiding unnecessary duplication of research work as well as suggesting further research through identifying any existing research gaps.

This chapter reviews the published literature that is related to the application of RFID technology in libraries and information centres. It specifically looks at how RFID technology has revolutionized library housekeeping activities and the challenges faced by libraries in adoption of the technology in a bid towards library automation.

2.2 Definition of RFID

RFID is an abbreviated form of **R**adio **F**requency **ID**entification. A number of definitions exist with all of them agreeing on the fact that RFID is a wireless form of transmitting digital data using radio waves between a stationery location and a movable object or between movable objects (Landt, 2005).

According to Garretson (2007) RFID is a technology that has quite a varied application in our day-to-day activities such as in the supply chain management, in tracking of property, as a security

measure among others, in restricting access, and recently in large sports events such as marathon to accurately provide identification and timing information of the competitors especially if they cross the finish line within a fraction of a second.

Fosso Wamba et al., (2006b) defined RFID technology as a wireless Automatic Identification and Data Capture (AIDC) technology which allows the accurate and automatic identification, tracking and of every product without human intervention.

Ward (2007) on the other hand defines RFID as a technology that incorporates the use of electromagnetic or electrostatic coupling in the radio frequency (RF) portion of the electromagnetic spectrum to uniquely identify an object, animal or a person.

RRID technology consists of three components namely antenna, transceiver and a transponder (tag) with some scientists combining the antenna and the transceiver in to a reader.

The antenna uses radio frequency waves to transmit a signal that activates the transponder and the tag now transmits data back to the antenna. The reader converts the radio waves reflected back from the RFID tag into digital information that can be passed on to computers that can make use of it. (Library Philosophy and Practice Vol. 8, No. 1, 2005).

In the library RFID is defined as a technology that allows the library information materials (including tapes which cannot be fitted with electromagnetic strips) to be fitted with RFID tags to automate material handling using a conveyor and a sorting system, to track, locate and manage library information materials, to allow users carry out self-check-in and check-out and enhance the security of library materials which reduces theft of resources.

2.3 The history of RFID.

It is generally said that RFID traces its roots back to the World War II with those countries involved using radar technology which was discovered by Sir Alexander Wattson-Watt in order to warn a country of an approaching plane while it was miles away. It was under Wattson-Watt that the British designed the IFF (Identify Friend or Foe) system where they put a transmitter on the plane which upon receiving signals from radar stations on the ground stations, it would then broadcast a signal back that identified the plane as friendly. In one incident during World War II, British Royal Air Force bomber crews made a security breach which resulted in Allied planes being shot down (Identification Friend or Foe IFF Systems: IFF Questions & Answers)

Ollivier, (1995) claims that it is the combination of radar technology and the IFF system that formed the beginning of RFID which was further developed in the late 20th century.

According to Landt (2001) the first publication on RFID was presented by Harry Stockman titled "Communication by means of reflected power" during the proceedings of IRE in 1948.

In 1950s and 1960s, various papers were presented which tried to explain how RF energy had the potential to remotely identify objects. Through the use of radio waves, companies started commercializing anti-theft systems to determine whether an item was paid for (Fusso Wamba et al., 2006). This was an era of exploration and development of RFID technology following the technical developments in radio and radar in 1930s and 1940s.

Major developments continued in the 1970s with developers, inventors, academic institutions actively working on RFID. In early 1973, Mario W. Cardullo claims to have received the first United States (U.S) patent for an active RFID tag while Charles Walton received a patent for a passive tag which was used to unlock a door without a key in late 1973.

The major applications at this time were in animal tracking, vehicle tracking, factory automation, and access control and supply chain management. Full implementation was realized in 1980s and

1990s with over 3 million rail cars in N. America being tagged in 1990s and more books begun to appear which were devoted specifically to RFID in the late 90s (Finkelzeller, 2003). It was in 1990 when Ultra-High Frequency (UHF) was introduced which offered long read range with a faster rate of transferring data. The conventional 125KHz frequency commonly referred to as low frequency (LF) was moved up to the radio frequency to a higher frequency of 13.6MHz which increased the application of RFID in areas such as in payment systems, in contactless smart cards, in anti-theft systems and in trade control and in libraries (Kern, 1999) and today the frequency has gone up to 2.5 MHz.

The tags which were being produced at this time had improved with a reduction in size and improved functionality such as the use of low voltage, long read range, fast data transfer rate and could be used in harsh environments.

2.4 RFID in libraries and library automation

A library houses information materials that are meant for the library clients to use. The 21st century library not only provides the resources on the library shelves but also targets to enhance efficiency in the retrieval and provision of access. It has becomes widely known that knowledge is power and the process of knowledge acquisition ought to be efficient- less time consuming, flexible and using minimal efforts. It is on this notch that library automation is critical- to improve the quality of service to the library users by ensuring that the five principles of librarianship as outlined by S. Ranganathan are followed to the later.

Library automation is the adoption of ICT in libraries where computers are used to enhance the basic activities of the library.

Library automation has developed over time in the different functions of the library. The period from 1930-1960 marked the beginning of library automation where computers were used to automate library circulation and acquisition while the period between 1960s and 1970s saw improvements in the use of computers and this led to an increase in library automation (Meckler, 1991). Since then, libraries have increasingly continued to automate their major processes and services such as cataloguing, acquisition, and circulation of library information materials (Computer-Based National Information Systems: Technology and Public issues, 1981:8).

Today, barcodes and electromagnetic strips are the technologies currently in use in most of the university libraries in Kenya in automating circulation and security management.

Library automation had begun to shifting slowly from the conventional barcode technology to RFID technology in the late 1990s. This new automation technology (RFID) integrated the barcodes, the scanners/barcode readers the book cards and the electromagnetic strips into a RFID tag (Chachra, 2003) which made the storage of bibliographic records, the circulation, the tracing of the location and ensuring the security of the information resources possible.

The first country to introduce RFID technology in its public library was Singapore while the Rockfeller University Library might have been the first university library in the U.S to adopt RFID. The adoption of this technology in developing countries has been slow due to the cost involved and the size of the library collections. In Kenya, CUEA library was among the first libraries to adopt RFID technology back in 2013.

2.5 Application of RFID in libraries.

It is a technology that has come to solve a number of issues in the library ranging from security of library materials, enhancing customer service, making shelving fast and efficient, enhancing circulation of materials among other issues.

2.5.1 Security management.

Libraries spend a lot of money and time in acquiring their collection, some of which are priceless and irreplaceable. Theft and loss of these resources can be devastating for any library. Information resources can be stolen by ill-mannered and selfish patrons who take them out of the library to use them at home or elsewhere and/or sell them. Theft of information resources has been in existence since the advent of libraries. In fact, the ancient libraries used chains to tie the books on the tables to prevent users from stealing them. Ancient librarians also adopted what was referred to as closed shelf system where users could not access the shelves by themselves: they had to be assisted by the librarians who would go to the shelves, retrieve the information resources and hand them over to the users. This was quite inefficient and very tiresome and it did not empower the user by giving them the knowledge to access the resources without help. It also infringed on the privacy of the readers since the librarian had to know which reader read which information resource.

Bordens (1996) carried out a study to determine the extent of library security problems and he established that library collection was at a high risk of theft and he proposed the need for adoption of security measures that would solve this problem. According to Bregel (2010), in any academic library with average security measures in place, the rate of losing books stood at around 3% from its collection per year.

The elimination of book theft in a library is unachievable since most security measures are vulnerable to misleading the system and some users have also learnt ways of defeating the systems. There are various reasons as to why books are stolen such as the pressure of academic work load (Weiss, 1981), strict library policies and procedures by the library (Boss, 1995), internet-based book buyers such as e-bay.com, amazon.com which provide a fixed market of stolen books (SCONUL, 2003).

There are different security measures adopted by libraries which can be broadly categorized into physical and electronic surveillance systems. The physical measures of ensuring that users did not steal library resources was adopted such as inspection of bags and frisking of patrons as they left the library, leaving bags as patrons enter the library, having secured window which minimizes the chances of passing books through the window. Libraries have also adopted the mechanism of manning the entrance and exit. Most librarians are, however, more popular with the installation of security gates in libraries to minimize theft of resources (Boss 1995).

The electronic surveillance systems (EAS) have also come in handy to help in ensuring the collection of the library is secure. With the adoption of computers in libraries, the barcode technology and the electronic magnetic strip technologies were introduced to solve this problem. The barcode is useful in checking-in and checking-out of the material while the electromagnetic strip is a security measure. The EM security gate automatically scans the items carried by the patron to identify the magnetic status of the strip attached to the book. If the strip is magnetized, then the item has not been properly checked-out and this gives a warning by activating an alarm sound (Kern, 2004). However, there still exists some loopholes with the barcodes such as the need for in-sight relationship between the reader /scanner and the barcode on the item, repetitive related injuries, some barcodes which may not read, time consuming and short read range (Lindquist, 2003). Library personnel can influence theft in the library untrained staff can be responsible for most of the library security problems and called for the need to have well trained and professional library staff. In another study carried out by Bahr (1989) on internal theft done by library patrons and staff, he argues that library staff could easily defeat the library security system and this type of theft is the hardest type of theft to prevent since the staff are closely involved in its implementation and called for more and wider research on a technology that could offset these

challenges.

RFID is the technology that has been widely adopted to offset these challenges. It combines the barcodes, the scanners, the book cards, and the electromagnetic strip into a tag. The bibliographic information about the information material is kept in the tag which is read by the RFID readers without the need to align the item with the reader. Once read, the item is deactivated and checked out to the patron carrying the item. At the exit, security gates, fitted with security alarms and manned by security guards are fitted which help to ensure that no un-discharged item leaves the library. According to Palmer, M. (2006) and Mehrjerdi, Y. Z. (2011) the adoption of RFID reduces theft of information resources in a library.

2.5.2 Circulation

It is at the circulation desk where a patron largely rates the quality of service a library offers. Long queues especially at peak hour of the day are very discouraging to users and lead to low quality service and subsequently user dissatisfaction. The time taken to check-out or check-in items and the procedure involved is a major determinant of patron satisfaction. RFID technology offers a better functionality compared to barcode readers as it can check-out several books at once unlike barcodes where each item must be scanned separately (Pandian, 2010) and within a line of sight (Finkelzeller, 2003).

Once the RFID tags (which operate automatically) are embedded on the information resource and on the patron cards, the RFID readers are used to read the tags and update the information on library management system (LMS).

Users can carry out self-check-out and check-in on a 24/7 basis using RFID technology which helps to free the library staff time for other service to users. The library can install RFID kiosks

outside the library or even at busy points such as at the airport, bus stop or restaurants where patrons can return the borrowed resources at the point of their convenience. In Singapore, all public libraries have installed RFID and patrons can return borrowed resources at any library conveniently which makes library services very flexible to the users. Library users do not have to schedule to visit the library before it is closed or during the official working hours or working days of the week.

The circulation librarian is hence saved a lot of time which can be used in doing other library activities. There is a notable reduction in the handling of the information resources by the circulation librarian and this reduces the cases of repetitive motion injuries.

Once a user with a RFID tag enters the library, the details are read by RFID tags and displayed on a computer. Upon leaving the library with tagged information resources, they are automatically checked out in the user's account. Multiple items can be read by the RFID tags within a fraction of a second and without the need to align each item to the readers as it is the case with barcodes. No item can leave the library without being checked out properly if the RFID system is operational and well maintained.

This technology gives users a high level of comfort when checking in and checking out information resources. With a properly integrated RFID system, users can be able to make electronic payment for library overdue fines and other library services such as photocopy, printing, membership fee among others (Kavulya, 2014, Renold & Rani, 2013).

2.5.3 Shelving.

In any library, the information materials need to be stored in a manner that allows users to retrieve them within the shortest time possible. To ensure this, there are classification schemes which allow books to be given call numbers which determine the location of the items in relation to other items. Most academic libraries in Kenya use the Library of Congress (LC) classification scheme to classify their information materials. The items are then placed in order on the shelves for the users to retrieve them.

However, the process of shelving is quite time consuming for the library staff and some end up mis-shelving the items due to the tiresome nature of the process or due to the pile of activities that the librarian has to complete within a certain duration. It becomes quite frustrating for users who cannot identify the location of the books on the shelves due to mis-shelving.

RFID technology has been adopted to solve this problem in helping the library staff to save quite a lot of time for other constructive activities and saving the time of the user as well as improving the quality of the library services.

With the use of sorting machines and conveyer belts, RFID technology can help the library in sorting the checked in books automatically into separate bins the library staff can pick and comfortably place on the shelves.

The staff can also use hand-held readers to check if all items within a shelf are correctly shelved reducing the chances of having books that are on the wrong shelves (Ahmad et al., 2011).

2.5.4 Information resource retrieval

It is important that the resources of a library be easily retrieved by the users from the shelves. In a library with thousands of collections, this can be time consuming and hence the need to ensure that the retrieval process is effective. Ensuring that materials are well shelved helps to achieve this. Rathinasabapathy (2013), in his study on the application of RFID in libraries identifies the use of RFID technology in libraries as a major milestone in ensuring that patrons retrieve resources within a short time. RFID technology allows for users to use hand held scanners are also provided for the users who can use them to identify a certain book within a fraction of a second (Kavulya, 2014). It is a technology that really helps in achieving the fifth law of librarianship- save the time of the user.

2.6 Advantages of RFID

The use of RFID in libraries to a great extent changes the quality of service to the users, the time they take to access a certain library service, the procedure followed, it also saves the library resources such as reducing replacement costs due to theft of information materials, fewer personnel for the library, and saving the library staff from repetitive stress related injuries and saving their time (Rodriquez-Silva et al,. 2007).

Library users can check out information resources on their own without the need for queues at the circulation. According to Howard & Anderson (2007) one way of reducing queues at the circulation is through the adoption of RFID. To return the borrowed items, users can use the self-check-in kiosks (machines) to return them even when the library is closed, over the weekend or even very early in the morning. With a well-integrated system, items can be returned in any library or in check-in kiosks which can be located at busy points (Kapoor et al., 2014). All this increases

the flexibility of the library services which improves user satisfaction.

RFID reduces theft cases in a library due to the improved material security. The tags which carry the bibliographic information about the can activate an alarm at the exit gate if the item has not been discharged out properly. The reduction in theft cases in a library is good news as the resources that could have been used to replace the items is directed in other needs of the library.

2.7 Disadvantages/challenges to implementation of RFID

The adoption of RFID has offered a number of challenges to libraries such as the high cost involved which is said to be the major disadvantage of RFID technology. It is one of the major factors influencing acceptance of this technology, although the production costs of RFID have reduced significantly (Collins, 2004). However, the retail price is far higher especially for library collections with over 100,000 items, which makes it expensive for them to implement RFID technology (Blutters, 2008). Complementary costs such as the costs of the readers, and other peripheral equipment and application software is a huge burden to the budget of the libraries which operate under limited budgets.

Another disadvantage of this technology is the ease at which it can be compromised. According to Boss (2009) it is possible to compromise an RFID system by wrapping a few layers aluminium so that the foil blocks the radio signals from reaching the tag or by placing two or more information materials against one another so that one tag overlays another.

The removal of exposed RFID tags which are not always concealed in the information material is another serious disadvantage facing RFID technology in the library.

This not only poses a cost related challenges but also a loophole in the security of the library

collection (Selamat, & Majlis, 2006).

Despite the common solution that have been adapted by libraries to solve this, nothing much has been achieved to solving this. Libraries started imprinting the RFID tags with their logo in a bid to make them appear to be bookplates, or putting a printed cover label over each tag but the removal of the tags (Haley and Lewis 2004).

METHODOLOGY

3.1 Introduction

In this chapter, the researcher explains and justifies the methods and procedures that were used to acquire primary data for the study. It describes the research design to be used, location of the study, target population, sampling techniques, and research instruments. Finally, the chapter discusses data collection techniques, data analysis and ethical issues considered during the data gathering and analysis process.

3.2 Research design

A research design according to Orodho (2003) is a scheme, an outline or a plan that once executed, it generates answers to the research questions of a study. Case study design was chosen as the most appropriate research design methodology. This is due to the fact that the study is based on a single case which is the CUEA library. According to Abdulhadi (2003) and Goldhor (1972) case study is strongly advocated for in conducting research in the fields of library and information science. Restricting the study to only a single case enables the researcher to study the case in detail. Other research designs are inappropriate for instance, survey research design which advocates for collection of data from a number of cases but this study is based only on a single case which is the CUEA library and surveying a number of academic libraries to study on the adoption of RFID will also be very costly in terms of time and traveling expenses. Experimental design on the other hand is totally inappropriate since no experiments will be conducted.

3.3 Location of the study

The study is to be conducted in the CUEA library. This library is located in CUEA main campus in Karen, Nairobi County housed by the university's five floor Learning Resource Centre (LRC)

building which comprises the 3,000 seater ultra-modern library which has installed RFID technology, a 1200 seater state-of-the-art e-conferencing facility, a 350 seater cafeteria, the ICT centre, the multimedia centre, the electronic publishing centre, the language laboratory, the electronic museum and archive, bookshop and curriculum development centre ("The Library of the CUEA- A Bastion of Knowledge", 2016).

In the library it is possible to find the target population which comprises of staff (librarians, senior library assistants and library assistants) and users/patrons (undergraduate students, post graduate students, teaching and non-teaching staff). This library carries out the basic information processing services including acquisition, cataloguing, research/writing consultancy services, referencing, information literacy services, circulation, ensuring security of the information materials, provision of IT services, and reading space among other services. However, the research will focus on the circulation of information materials, the security management, library stock management, and shelf management using RFID technology. From all these categories of users and library housekeeping activities, the researcher's representative sample will easily be attainable.

3.4 Target population

A population in research is defined as an entire group of individuals, events or objects that share common observable characteristics (Mugenda and Mugenda, 1999). The target population of the study comprised of 35 people. The target population is heterogeneous.

It will comprise of the circulation librarian, the library attendants who are involved in shelving and library stock control, the library security guards at the exit, and the library users. The various categories of people that are targeted by the researcher and their numbers are as indicated in the table below.

Table 1.1 Target population

Category of respondents	**Number/value**
Circulation librarians	5
Library attendants	6
Library security guards	4
Library users	20
TOTAL	35

The library's top management such as the university librarian and his deputy will not be included in the target population. This is because they are likely to interfere with the integrity of the data collected because of their administrative position. They may want to paint a good picture of the adoption of the RFID system.

Administrators usually don't gather management information from the subordinate staff they are managing but instead from other administrators that are above them or the same level as them. They are therefore separated from their subordinate staff and hence likely to give information totally in contrast with the other staff or that is meant to bring the perception that they are doing extremely well.

Other librarians such as the circulation librarian, library attendants, the library security guards and the user will be involved in the study since they are the ones who deal directly with the RFID system in their day-to-day activities since the study wanted to gather their views regarding various aspects of this technology in the library.

The circulation librarian oversees the process of lending out and receiving items from users, the library attendants are the ones responsible for shelving and inventory management, security guards

at the exits respond to the alarm raised by the RFID system if an item is being taken out of the library without being properly discharged, while the users are involved in the processing of borrowing and returning borrowed items. All these groups of people are therefore involved in the basic library activities which enables them to have a better say regarding new RFID technology adopted by the library a few years ago which replaced the barcode technology.

3.5 Sampling techniques and sample size

A sample is a small subset of the target population that has been chosen for study and it should represent the population fully for sufficient statistical analysis to be done (Mugenda, 1999) while Orodho and Kombo, (2002) defines a sample as a finite part of a statistical population whose properties are studied to gain information about the whole population. Sampling is defined as the process of selecting a small number of respondents from the population for research in such a way that those who have been chosen will be good source of the required information for the study in accordance with the research questions (Gay et al., 2009).

3.5.1 Sampling techniques

The researcher will use random sampling method to determine the sample size. All the different categories of respondents in the target population will separately go through random sampling from which a representative and proportional sample would be achieved.

This method is non discriminative in that it provides an equal opportunity for every individual in the target population to be included in the sample (Orodho and Kombo, 2002). It was also preferred over other methods such as snow ball technique since the target population could easily be identified.

The study assumed that the information to be collected from the sample represents the views of the target population and the position on the ground related to the RFID system.

Some of the reasons why sampling is preferred included: Collecting information from only a section of the target population saves on time. Less time is spent gathering information as opposed to if the entire population is to be contacted.

It also saves on costs since the researcher will just be required to produce fewer questionnaires to be distributed among the respondents.

3.5.2 Sampling size

From the target population of 35, a sample size of 30 was used. The table below shows how the different categories of respondents were represented so as to ensure the sample was both representative and proportional.

Table 3.1 Sample size

CATEGORY OF RESPONDENTS	TARGET NUMBER	SAMPLE SIZE
Circulation librarians	5	4
Library attendants	6	5
Library security guards	4	3
Library users	20	18
TOTAL	**35**	**30**

Using a representative sample ensures that a more accurate information is collected since the sample size includes all categories of people to be investigated are represented and in the right proportions.

3.6 Research instruments

The study will use questionnaires to gather information about the research topic. To meet the research objectives, the researcher has prepared two sets of questionnaires. One set is meant to collect data from the library staff (appendix IV) concerning circulation, shelving, stock management and security of the information materials while the other set is designed to collect data from the library users.

The questionnaire is designed into two main sections. There is background information section (Section A) inquiring information about the respondent. This regards information on one's job title, period worked in that library, knowledge about the RFID technology. The period staff members have worked in this library will help gauge their level of familiarity with the RFID system.

Section B is designed to gather data about the use and adoption of RFID technology in terms of understanding RFID, its predecessor (the barcode) technology, challenges encountered when using barcode, the solutions offered by RFID and the challenges encountered when using the current RFID technology in the library.

The other category of the questionnaires will be meant for the library users and it will aim at collecting data from the users of this technology. This will majorly be to avoid bias from the librarians who might want to portray the system as flawless. It has two sections also with section A collecting background data about the user such as their year of study and if they saw the introduction of the new technology. Section B targets collecting data on the users' understanding of using this technology, if they were trained, their basic uses when in the library and any inconveniences encountered if any.

Some of the reasons behind using questionnaires as the most appropriate instrument included the

fact that they can be made anonymous. Nobody is required to put his/her name on the questionnaire. Making it anonymous enables the respondents to respond to it without fear of being noticed.

The questionnaire is also flexible. The respondent can answer it at their own free time. Answering the questions when not under pressure enables one to provide more accurate views about the matter under study. This is unlike the interview where both the respondent and the researcher will be required to be present during the interview. The interview method would also make the interviewee to be under pressure to respond to the questions and may end up providing responses with the aim of just completing the interview.

It is also time saving since the researcher can reach out to many respondents and collect a lot of information within a very short period of time. This is unlike in interviews where the researcher will be required to interview one person at a time ending up spending more time on few interviewees and collecting less information.

To ensure that the data collected was representative, the questionnaires will be randomly distributed to all the required service points and in the right proportion and considering gender.

3.7 Data collection techniques

The researcher will randomly distribute the questionnaires among the sampled population in the right proportions. Questionnaires will be first distributed to the library staff and then the users. The respondents will be given enough time to respond to the questions after which they will be picked back by the researcher in the order in which they were distributed. Most of the questions just required the respondents to tick where appropriate in all the sections.

3.8 Data analysis

Data analysis is the process of examining the data that has been collected during the study and making deductions and inferences (Kombo and Tromp, 2006). It is a process that involves a number of activities such as drawing up a summary of the raw data, categorizing the raw data, rearranging and ordering in order to be able to draw conclusions and make decisions based on the collected data (Mbwesa, 2006). The study will use qualitative approach to analyze data. Themes will be used to analyze the data collected. The themes used will be based on the application of RFID in the library which included: check-in and check-out, security management, shelving and retrieval, stock control. Analyzing the data by use of themes brings out a more clear picture of how these different elements RFID technology and which element has the highest score, which one the lowest and which ones moderate. This kind of information would enable the academic libraries to know state of this technology and to know which areas to improve the system and user satisfaction. The findings of the study will be tallied and represented in tables. Tallying provides a faster way of representing the data in a table. Use of tables is the most appropriate method since it is able to represent each aspect separately and indicate its level of respondents' satisfaction. Conducting data cleaning too ensures that only completely filled questionnaires were tallied. Data will also be analysed using percentages, pie charts and bar graphs. The data gathered will not be statistical hence qualitative approach was the best method to use to analyze it.

3.9 Ethical considerations

The study will take into consideration ethical issues both during data collection and analysis. An introduction letter requesting for permission to collect research data will be presented to the various categories of library staff and library patrons that make the target population before collecting data from them.

During data collection, respondents' rights regarding their participation in the study will be clearly spelt out in the introduction letter. The respondents have the right to choose whether to or not to participate in the study, right not to answer questions they are uncomfortable with but explaining to them the importance of attempting all questions, right to know why the research will be done.

To ensure respondents` anonymity, they were not required to write their name anywhere on the questionnaire.

Data integrity was also ensured while analyzing the data collected. Data honesty will be ensured by reporting the findings exactly as they were and by not manipulating them in any way. Respondents' privacy will be ensured by not disclosing their identity and also generalizing the findings. Data confidentiality will be ensured by only sharing the information with the intended individuals.

DATA ANALYSIS AND PRESENTATION

4.1 Introduction

This chapter analyses, presents and interprets the data collected during the study. The focus is on the adoption of RFID technology in libraries in respect to the respondent's responses on circulation, security management, shelving, information retrieval and any inconveniences encountered when using this technology. It also includes a section of the respondent's background information which aims at collecting data from the respondents especially on their designation and the period they have been working in the library.

4.2 General analysis

The research had two main categories of respondents which included the CUEA library staff and the users of this library. The data that was collected from them was analysed using tables, percentages, pie-charts and graphs. Out of the targeted population of 35 respondents, a sample size of 30 was attained. The study sample comprised of 12 library staff and 18 library users. The library staff who formed the sample were of different designation including four circulation librarians, five library attendants and three library security guards. The sample of this study is summarized in **figure 4.1** below.

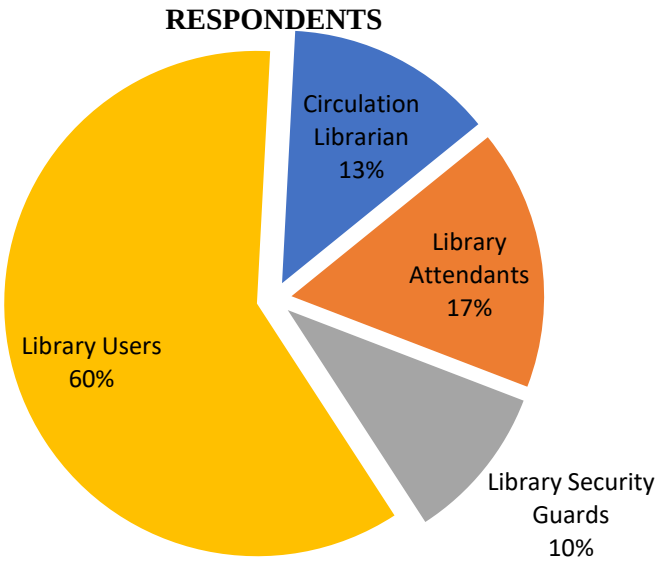

Figure 4.1: A summary of the sample size

The researcher distributed 30 questionnaires and 25 of them were returned representing 83.3% return rate which was considered suitable for the study. The returned questionnaires had a good response rate with only a few respondents avoiding some questions though this did not affect the final analysis. The table below summarizes how the distributed questionnaires were returned by the different categories of the respondents.

CATEGORY OF RESPONDENTS	QUESTIONNAIRES DISTRIBUTED	QUESTIONNAIRES RETURNED
Circulation librarians	4	4
Library attendants	5	4
Library security guards	3	1
Library users	18	16
TOTAL	**30**	**25**

Table 4.1: A summary of the distributed questionnaires and their return rate

The four questionnaires which were distributed to the circulation librarians were all returned, while

four out of the five questionnaires distributed to library attendants were returned representing 80%, only one out of the three questionnaires issued to the library security guards were returned representing 33.3% return rate but this did not affect the study significantly. The library users recorded the second highest return rate of 88.9% with 16 out of the 18 questionnaires being returned.

The following table provides a summary of how the library users responded to the questions on their use of RFID technology in the library.

QUESTION	**YES**	**NO**
Do you understand what is RFID	16	0
Have you been trained on how to use RFID in the library	14	2
Do you understand the use of security gates	16	0
Have you ever been inconvenienced by RFID system	3	13
Is RFID more convenient that being served at circulation counter	16	0
Are the librarians well trained to use RFID	12	4

Table 4.2: A summary of users' response to the questionnaire. (Appendix III)

From the above table, all the respondents responded positively to their understanding of the RFID system in the library which showed that they were oriented properly when the new system was installed though 2 of them felt that that they were not trained properly to use the RFID system in the library. Training users of an information centre is a way of empowering them and it helps the library to save on its resources such as the time that staff in directing and assisting users. This time can be utilized in other productive activities within the library. Users who are trained are likely to

make fewer mistakes hence reducing the chances of the library incurring costs arising from mistakes of the users. Empowering customers through training is also a way of giving users a sense of belonging and trust from the library which increases their satisfaction in the library.

All respondents responded positively on the use of the security gates at the exit of the library. The security gates at the exit, activates an alarm if materials have not been properly discharged and the respondents seemed to be aware of the importance of these gates.

Only three out of the 16 respondents said to have been inconvenienced by the RFID system. All of them described the inconvenience as power related issues. The library has a standby generator to supply power when there is an issue with the mains. One of them stated that sometimes the readers fail to read the tags which results to activation of the alarm at the exit gate.

It was overwhelming how all respondents positively responded to if RFID system was more convenient in checking in and checking out materials than being served at the circulation counter. The short time taken, the flexibility and the privacy involved were attributed to this.

On the librarians' knowledge and understanding of the RFID system, 12 out of the 16 users felt that the librarians were well trained which represented 75% of the users. As urged by Bahri & Amir (2013) it is important that users have confidence with the librarians to improve their level of satisfaction. This is possible through training the librarians extensively though this can be costly for the library, it really pays (Blutters, 2008)

The library staff on the other hand, responded to the questions as shown in the following table.

QUESTION	YES	NO
Do you understand what is RFID	9	0
Have you used the barcode technology before	9	0
Did you encounter challenges with the barcode	9	0
Do you think RFID has solved the barcode problems in your library?	8	1
Do you think the adoption of RFID in the CUEA library has lead to better services to the users?	9	0
Would you encourage other libraries in Kenya to adopt this technology?	7	2

Table 4.3: A summary of library staff response to the questionnaires. (Appendix IV)

Table 4.3 above shows how the library staff (circulation librarian, library attendants and security librarians) responded to the questionnaire. All of them respondent positively to if to if they understood RFID system in the library, whether they had used the barcode technology before and if they experienced challenges using the barcode technology. All the library staff also agreed that the adoption of RFID technology in CUEA library had led to better service to the users. On whether the adoption of the new technology had solved the basic library house-keeping activities, 88.9% agreed with the remaining percent disagreeing. This was attributed to the challenges encountered when using RFID and also due to the fact that RFID does not fully offer solutions to all challenges facing today's libraries (Makori, 2013) and the challenges involved in implementation (Kapoor, 2014). On nationwide adoption by other libraries, only 77.8% would encourage other libraries to

adopt this technology with 22.2% responding negatively. This was attributed to the costs involved in implementation (Kavulya, 2014), the resistance from staff who fear loss of their jobs (Kern, 1990), lack of standards (Rathinasabapathy, 2005) and technological challenges (Selamat, 2006).

4.3 Analysis on security management:

There was a question touching on security on both the users and the library staff questionnaire which aimed at collecting data on if RFID has had an impact on the security of the library materials. All library users who participated and returned their questionnaires confirmed to understand the use of RFID security gates at the exit and all library staff on blamed the barcode system for having loopholes in security management. This analysis concurs with the finding of Hayer and Lewis on their research carried out in Chicago Voyager Library (Hayer and Lewis, 2004). It is evident that the new RFID technology has revolutionized the issues of security of the CUEA library collection.

4.4 Analysis on circulation, shelving and retrieval.

The data collected from the library users showed that all of them used the RFID system to carry out self-check-out and check in of library materials. The library has installed self-check-out and check-in kiosks which are quite popular among the users. All the library staff responded positively that RFID solves the problems of shelving. Contrary to the barcode technology where shelving was tedious, time consuming and lead to mis-shelved item, they all responded that RFID has solved these problems. Eight out of the 16 users, representing 60% confirmed to have used the technology in retrieving resources from the shelves.

4.5 Analysis on challenges and inconveniences

Both the users and the library staff recorded a variety of challenges that are experienced when using this technology in the library. The major challenge noted by 18 out of the 25 respondents which represented 72% was power surges. Though the library has a standby generator, it took time

for the reboot process of the RFID system to take place which not only inconvenienced users but also posed a security threat to the materials. Eight out of the nine library staff which represented 88.9% who returned their questionnaires stated that another challenge was on staff reduction due to the fact that the system could do some of the duties that were previously done by librarians.

4.6 Analysis on the rating of RFID in the library

The respondents were requested to rate RFID technology in enhancing the basic activities of an scale of 1-5 where 1 represented very poor, 2 represented poor, 3 represented good, 4 represented very good while 5 represented excellent. The results are analysed on the graph below

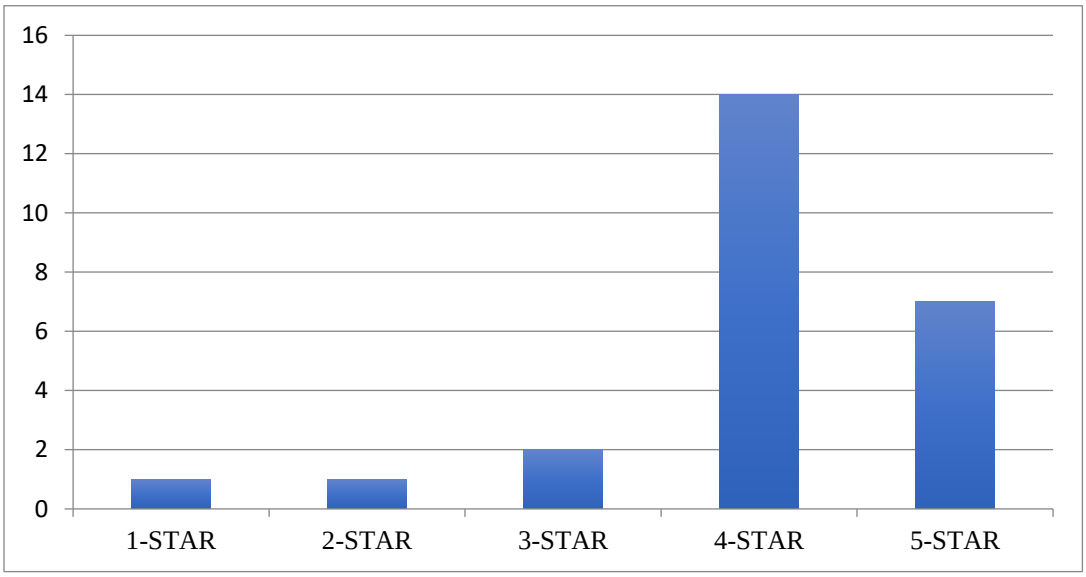

Figure 4.2: A summary of the rating by respondents on a scale of 1-5 star.

The above graph show that 4% rated the technology as 1-star, another 4% as 2-star and 8% rated it as 3-star, 56% rated it as 4-star while 28% rated it as 5-star. This information is represented in a pie-chart below.

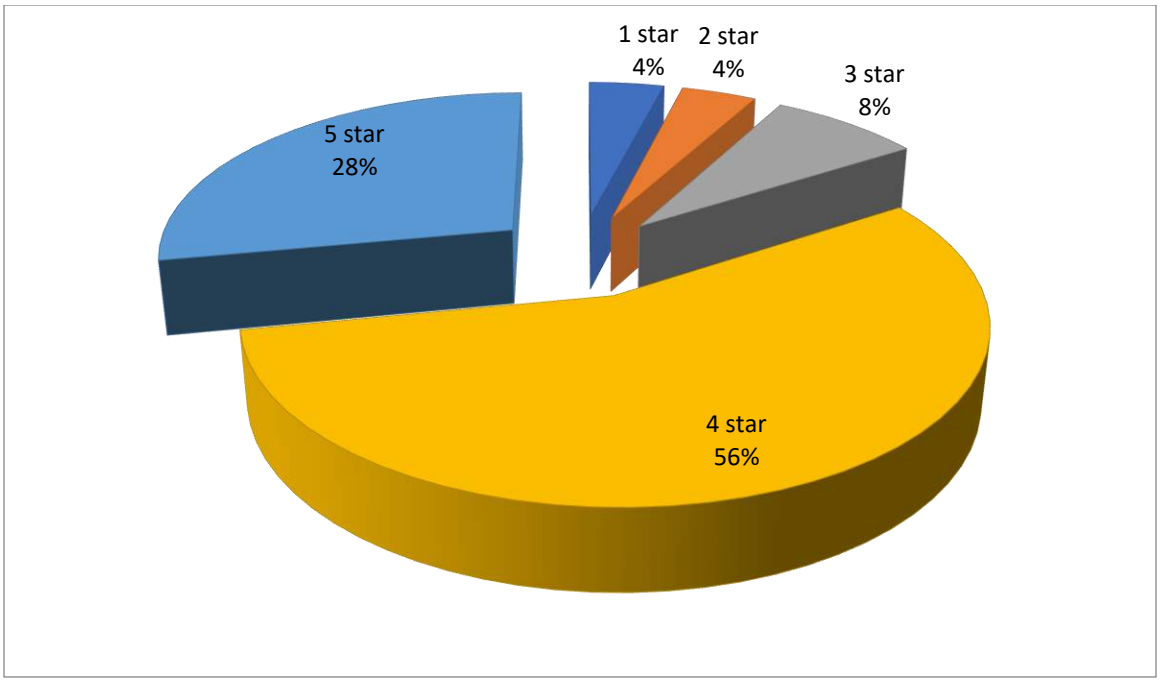

Figure 4.3: A pie chart showing the rating of RFID by respondents

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents a summary of the study findings, conclusion and recommendations that may be taken into account by librarians and information professionals considering the adoption of RFID technology in their libraries. Also stated are recommendations for further research to improve the knowledge base and research gaps that have not been handled in this study.

5.2 Summary of the findings

The summary of the findings of this study were based on the research objectives and research questions stated in chapter one and as presented chapter four of this research. These included the following:

i. Security management.
ii. Circulation.
iii. Shelving.
iv. Information retrieval.
v. Challenges and inconveniences.
vi. Overall rating of RFID in libraries.

5.2.1 Security management

The findings showed that RFID technology has improved the security of information materials in the library. The library staff noted reduction in theft of information materials due to the installation of the security gates and the RFID system which is not easy to manipulate since information is read wirelessly. The awareness of how RFID security works amongst the users showed that they were aware of the role it plays in ensuring information material safety.

5.2.2 Circulation

From the research findings, it was evident that all respondents use the self check-out and check-in stations to circulate materials. Circulation librarians were freed quite a lot of time which they would engage in other activities, they no longer handled books when being borrowed or returned. The injuries related to the process of opening books to read the barcode and the repetitive process of scanning barcodes, some of which never detected the signals was no more in the library. Users no longer have to queue at the circulation counter to borrow or return borrowed library materials which was stated as one of the major reasons why the system was rated 4-star by more than half the respondents of this study.

5.2.3 Shelving

The assistant librarians are involved in shelving and ensuring that information materials are well placed on the shelves and easily retrieved by users. According to the findings, shelving was no longer tedious, time consuming and inefficient after the adoption of the RFID technology in the library. They did not have to raise quite voluminous books and place them on the shelves only to find that they had placed them on the wrong shelf. Repetitive stress injuries were reduced among the shelving librarians. Sorting was now being done automatically by the conveyor belt placing the books in separate bins. They could easily confirm if the books were well shelved by using a hand-held scanner, though they are few in the CUEA library. This reduced the chances of having mis-shelved books.

5.2.4 Information resource retrieval

The use of RFID technology in retrieving resources from the shelves was low in the library it is time consuming to retrieve resources from the shelves for users manually and with RFID kit, this is catered for by provision of handheld scanners which produce an alarm sound when the

predetermined item is scanned from the shelves. In CUEA, the use of RFID hand-held scanners is low due to the fact that the scanners are quite few and cannot be provided to the large number of users retrieving items from the shelves. However, the study findings were that there is some use of RFID to retrieve resources.

5.2.5 Challenges and inconveniences

The study noted a number of challenges and inconveniences when using RFID in the library among users and the library staff. Most library staff cried foul of the cost related challenge. This is both in installation and maintenance. Power failure was also noted as another challenge which inconvenienced the users by having to stop them from exiting the library until the RFID system fully reboots which takes time. This is a security measure to ensure that no resources leave the library without being properly discharged. The library staff also mentioned that some users remove or tamper with the tags placed on the books. This has forced the library management to have security librarians at the exit to ensure that all items leaving the library have their tags intact.

5.3 Conclusion

Despite the challenges and inconveniences highlighted above, it is evident that RFID technology has a place in the library and information centres. It has revolutionized how the basic library activities are carried, how the 21st century libraries are built, how the library users access the library services and most importantly, it has also reshaped the skills of the 21st century librarians and other information professionals. RFID has solved major challenges that have been in the library for centuries such as theft of information materials, the long queues at the circulation counter. It has also offered solution to the challenge of struggling to pay overdue fines before the library doors are closed by offering an online payment system. RFID has proved worth adoption in the library by its flexibility in returning borrowed items and also in borrowing item through the self-check-

out kiosks.

Shelving and inventory management has never been as easy as it is with RFID. The conveyor belts help in sorting books automatically upon return and the shelving librarian simply pulls the bin to the shelves. They are also fitted with springs which allow books to be on the upper part of the bins saving the librarian the hustle of bending to pick the books from the bins. Stock control has been reduced to an activity of a few hours from what used to a week-long activity involving tens of library staff. With the use of a hand-held scanner, the shelf information can be sent to a database either with a wired or wireless connection.

Circulation counter is no more in RFID enabled libraries as there are self-check-out and check-in kiosks which allow users to borrow or return the materials by themselves. Users now feel free to borrow any item of choice as no one knows which items they are borrowing- there is increased privacy. There is flexibility of returning the items that have been borrowed. It's a 24/7 hours service. RFID system is the system of today's library.

5.4 Recommendations

From the study, the following recommendations were made:

Academic libraries need to consider implementing RFID technology in libraries as the advantages of this technology far much out way the current barcode technology. It is a technology that can grant a library many benefits ranging from saving resources to enhancing the quality of service the library offers to its users. RFID technology ensures that the basic library activities are effectively handled with few errors and within short periods of time.

For those libraries which have adopted RFID technology, this research recommends that they engage in continuous training of both its staff and users' so as to be able to reap all the benefit of

this technology. Training helps to ensure that there are fewer mistakes arising from the staff and users as well as in ensuring that the technology is fully exploited. It's a way of empowering those involved.

5.5 Recommendations for further reading.

This research recommends that the following area need to be researched on RFID:

i. A cost benefits analysis of the implementation of RFID in small libraries.

ii. The standards for RFID systems in libraries.

iii. Tag collision as a challenge in library RFID tag reading.

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